

DS Lecture 1

Introduction

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DENMARK

A distributed system is one in which the failure of a computer you didn't even know existed can render your own computer unusable.

- Leslie Lamport



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The computer you did not know existed

Pets 'go hungry' after smart feeder goes offline

25 February



Owners of a device designed to release food for pets say their animals were left hungry during a week-long system failure.

What is it?



What is a distributed system?

A **distributed system** is one where hardware and software components in/on **networked computers** communicate and coordinate their activity only by **passing messages.**

What is it?



Question:

How far (in meters) are you from a distributed system right now?

What is it?

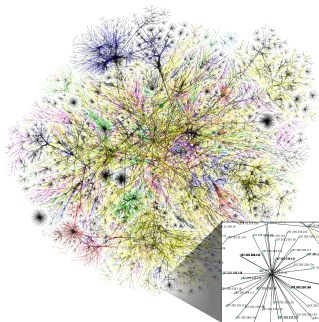


Figure: Visualization of the internet.

What is it?

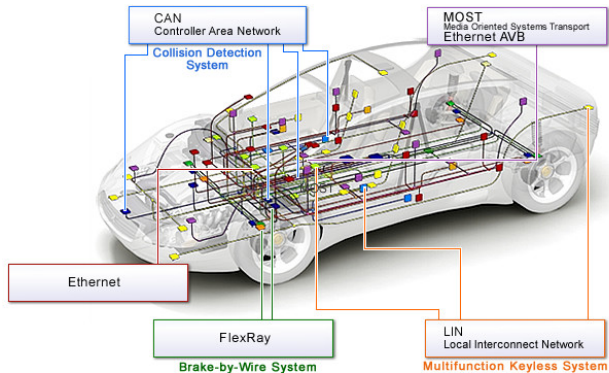


Figure: A (distributed) car

What is it?

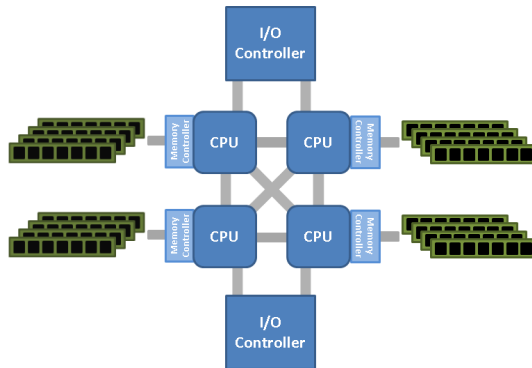


Figure: A modern CPU

Agenda



Consequences

Reasons for Distribution

- Independent Failure

- Amdahl's Law

Course Overview

Course Information

- Course Time Breakdown

- A note on Exercises

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ Non-determinism
- ▶ No shared state
 - ▶ Pass messages to synchronize
 - ▶ May not agree on time
- ▶ Everything can fail
 - ▶ Devices
 - ▶ Integrity of data
 - ▶ Network

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ **Non-determinism**
- ▶ No shared state
 - ▶ Pass messages to synchronize
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 - ▶ Network

Consequences

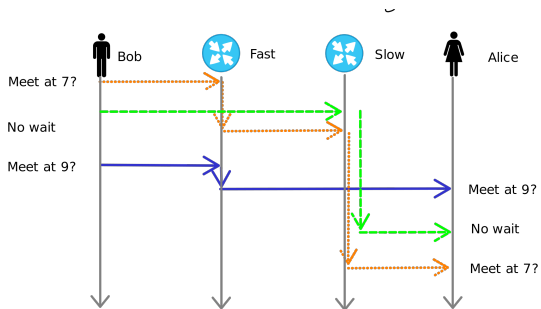


Figure: Properties of the network can lead to bad dinner plans.

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ Non-determinism
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Consequences

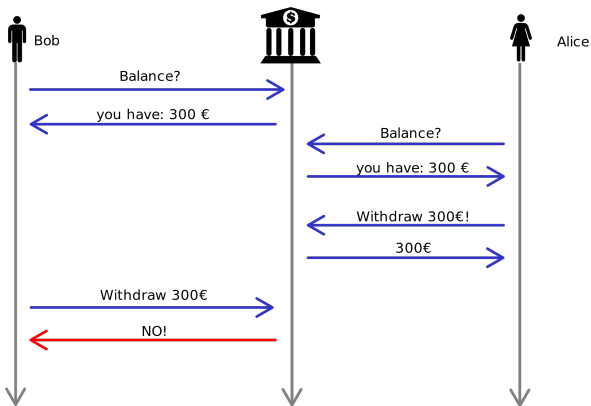


Figure: Lack of shared state can lead to unexpected behavior from the perspective of Bob.

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ Non-determinism
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 - ▶ **May not agree on time**
- ▶ Everything can fail
 - ▶ Devices
 - ▶ Integrity of data
 - ▶ Network

Consequences

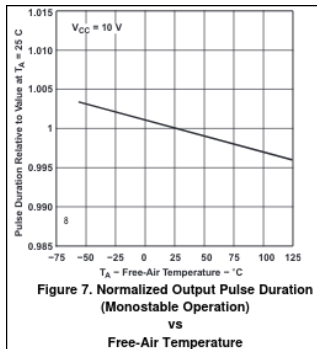


Figure: Measuring time is hard. (*Texas Instruments xx555PrecisionTimers*)

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ Non-determinism
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 - ▶ Pass messages to synchronize
 - ▶ May not agree on time
- ▶ Everything can fail
 - ▶ Devices
 - ▶ Integrity of data
 - ▶ [Network](#)

Consequences

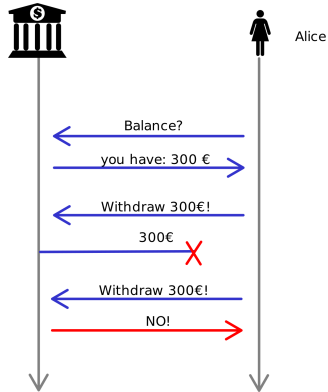


Figure: Failure of the network can lead to unexpected behavior from the perspective of Alice.

Consequences



- ▶ Concurrency
 - ▶ Deadlocks and livelocks
 - ▶ Non-determinism
- ▶ No shared state
 - ▶ Pass messages to synchronize
 - ▶ May not agree on time
- ▶ Everything can fail
 - ▶ Devices
 - ▶ Integrity of data
 - ▶ Network
 - ▶ Security!
 - ▶ Man-in-the-middle
 - ▶ Byzantine Failures

Agenda



Consequences

Reasons for Distribution

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- Amdahl's Law

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Reasons for Distribution

Domain

- ▶ Physical Limits
- ▶ Logical Limits

Redundancy

- ▶ Available
- ▶ Robust

Performance

- ▶ Economics
- ▶ Scalability

Reasons for Distribution

Domain

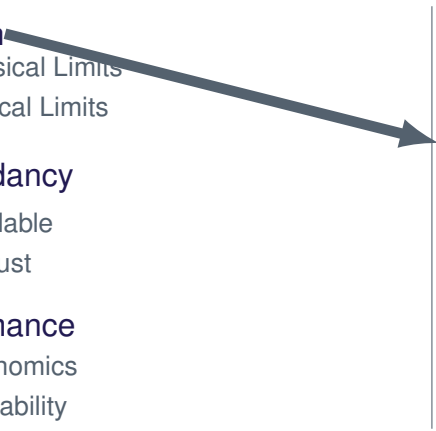
- ▶ Physical Limits
- ▶ Logical Limits

Redundancy

- ▶ Available
- ▶ Robust

Performance

- ▶ Economics
- ▶ Scalability

- 
- A diagram consisting of a vertical line. To the left of the line are three sections: "Domain", "Redundancy", and "Performance", each with a bulleted list. A large arrow points from the "Domain" section towards the vertical line. To the right of the line is a bulleted list of four items: "The internet", "Wireless Mesh Networks", "Industrial Systems", and "Ledgers (BitCoin, Ether)".
- ▶ The internet
 - ▶ Wireless Mesh Networks
 - ▶ Industrial Systems
 - ▶ Ledgers (BitCoin, Ether)

Reasons for Distribution

Domain

- ▶ Physical Limits
- ▶ Logical Limits

Redundancy

- ▶ Available
- ▶ Robust

Performance

- ▶ Economics
- ▶ Scalability

- 
- A thick grey arrow points from the "Redundancy" section header to a list of four items. A vertical line separates the left column of headers from the right column of items.
- ▶ 99.9% uptime
 - ▶ Backup
 - ▶ Database
 - ▶ Banking

Independent Failure

Redundancy

Dependent

One fail = system fail.

$$uptime = (1 - p)^N$$

Independent

All fail = system fail.

$$uptime = 1 - p^N$$

N (p=0.05)	Availability		Yearly Downtime	
	Dep	Ind	Dep	Ind
1	95%	95%	18 days	18 days
2				
3				
4				

Independent Failure

Redundancy

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N (p=0.05)	Availability		Yearly Downtime	
	Dep	Ind	Dep	Ind
1	95%	95%	18 days	18 days
2	90.25%		36 days	
3	85.74%		52 days	
4	81.85%		<u>68 days</u>	

Independent Failure

Redundancy

Dependent

One fail = system fail.

$$uptime = (1 - p)^N$$

Independent

All fail = system fail.

$$uptime = 1 - p^N$$

N (p=0.05)	Availability		Yearly Downtime	
	Dep	Ind	Dep	Ind
1	95%	95%	18 days	18 days
2	90.25%	99.75%	36 days	1 day
3	85.74%	99.99%	52 days	1 h
4	81.85%	99.999%	<u>68 days</u>	<u>3 min</u>

Independent Failure

Redundancy



Chaos Monkey is responsible for randomly terminating instances in production to ensure that engineers implement their services to be resilient to instance failures.

Figure: A Netflix approach to fault tolerance.

Reasons for Distribution

Domain

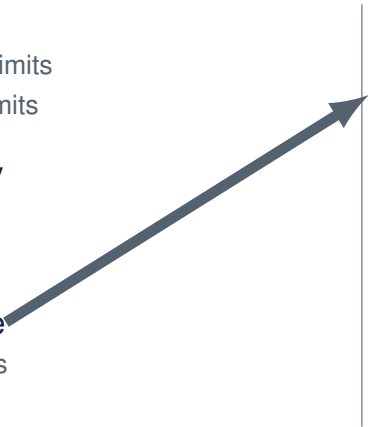
- ▶ Physical Limits
- ▶ Logical Limits

Redundancy

- ▶ Available
- ▶ Robust

Performance

- ▶ Economics
- ▶ Scalability



- ▶ Video Streaming
- ▶ Cloud Computing
- ▶ Supercomputers
 - ▶ Weather Forecast
 - ▶ Machine Learning
- ▶ Many inexpensive vs. few, expensive and specialized

Amdahl's Law

Scalability



F-Klub Accounting

- ▶ Each person does accounting for a month
- ▶ After all months are computed
 - ▶ Compute sum of year (1 person)
- ▶ 1 hour per sum computed

1 Person

12 Person



Amdahl's Law

Scalability

F-Klub Accounting

- ▶ Each person does accounting for a month
- ▶ After all months are computed
 - ▶ Compute sum of year (1 person)
- ▶ 1 hour per sum computed

1 Person

$$1 + 12 = 13 \text{ Hours}$$

12 Person

$$1 + 1 = 2 \text{ Hours}$$

$$\frac{13}{2} = 6.5 \times \textit{Speedup}$$



Amdahl's Law

Scalability

Amdahl's law

$$speedup = \frac{1}{1 - p + \frac{p}{n}}$$

p = fraction of parallel work

n = number of processors

Amdahl's Law

Scalability

Amdahl's law

$$speedup = \frac{1}{1 - p + \frac{p}{n}}$$

p = fraction of parallel work

n = number of processors

$$6.5 = \frac{1}{1 - (12/13) + \frac{12/13}{12}}$$



Amdahl's Law

Scalability

Amdahl's law

$$speedup = \frac{1}{1 - p + \frac{p}{n}}$$

p = fraction of parallel work

n = number of processors

$$6.5 = \frac{1}{1 - (12/13) + \frac{12/13}{12}}$$

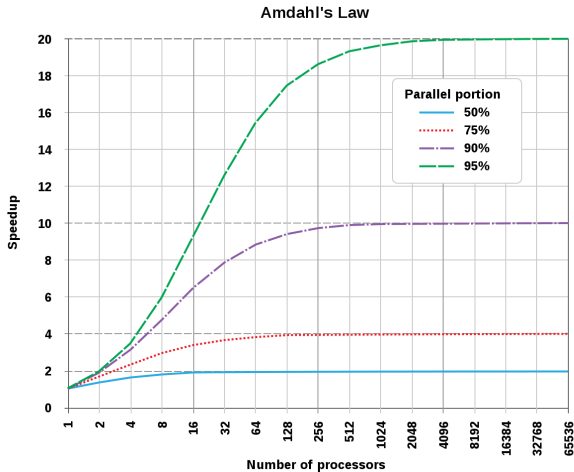
Minimize sequential part

Amdahl's Law

Scalability

Speeding up is hard

- Communication Overhead
- Locking and synchronization

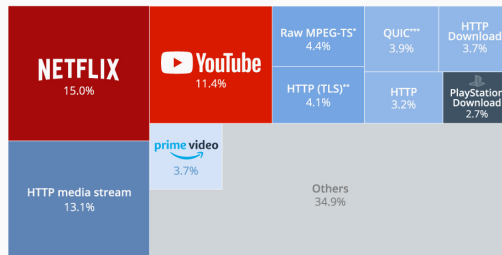


Examples

Scalability

Netflix is Responsible for 15% of Global Internet Traffic

Distribution of worldwide downstream traffic, by web application



* Digital container format for the transmission and storage of audio, video, and data.

** Security protocol

*** Network protocol designed to speed up online web applications



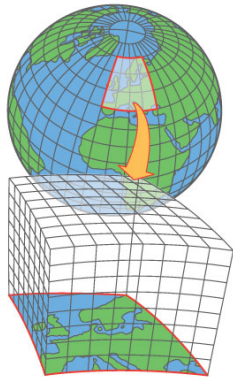
@StatistaCharts

Source: Sandvine | The Global Internet Phenomena Report

statista

Examples

Scalability



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Course Overview

Intro

1. Introduction

Theory & Algorithms

2. Models of DS
3. Time in DS
4. Multicast
5. Consensus
6. Distributed Mutual Exclusion

Application

7. Distributed Storage
8. Distributed Computing
9. Blockchains
10. Peer-to-Peer Networking
11. Internet of Things and Routing
12. Distributed Algorithms

Teachers: Michele and Elli

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Course Information



- ▶ 5 ECTS credits
 - ▶ 150 hours of work
- ▶ Written Examination
 - ▶ Internal Censor
- ▶ See moodle for details
- ▶ Read the study regulations



Course Time Breakdown

- ▶ 12 lectures of 2 hrs. = 24 hrs.
- ▶ 12 exercise sessions of 2 hrs = 24 hrs.
- ▶ 3.5 hrs. Reading per lecture = 42 hrs.
- ▶ Study exercise = 30 hrs.
- ▶ Exam preparation = 30 hrs.

Total

150 hrs

We reserved 13 lectures on the Calendar

Exam Information

- ▶ Written exam
 - ▶ currently, we plan for a 3 hours exam
 - ▶ a number of questions covering all the material taught
 - ▶ approximately 50% of questions on topics 2 to 6, and 50% to topics 7 to 12
- ▶ Exam aids
 - ▶ **one** sheet of A4 paper (2 sides) with notes
 - ▶ personal notes = must have been written individually by you and marked with your name and student code
 - ▶ you must submit the notes sheet together with the exam

A note on Exercises

Today:

- ▶ Warmup exercise to get you started
- ▶ Can be solved independently
 - ▶ ...but you are welcome to collaborate
- ▶ Find your TA and myself by email or DM in Teams
 - ▶ Nicholas Herrmann [<nhe@cs.aau.dk>](mailto:nhe@cs.aau.dk)
 - ▶ Michal Šedý [<medy@cs.aau.dk>](mailto:medy@cs.aau.dk)
 - ▶ Michele Albano [<mialb@cs.aau.dk>](mailto:mialb@cs.aau.dk)
 - ▶ Elli Anastasiadi [<ellia@cs.aau.dk>](mailto:ellia@cs.aau.dk)

A note on Exercises

Henceforth:

- ▶ Exercises in group-rooms
- ▶ We could not book a room for the exercises in Cassiopeia. We have group rooms at Frederik Bajers Vej 7D, 1. Floor.